Entropie
Relative Entropie von endlichen Quantenzuständen

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Abstract

This is the abstract.

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1 Section 1

1.1 Some Definitions

Question 1.1. Definieren hier die f-Divergenz allgemein für positive lineare Funktionale über von Neumann Algebren. Haben wir dadurch später Vorteile, oder willst du das einfach erstmal so allgemein wie möglich machen? Für die Physik und viele Resultate werden wir schätze ich mal Zustände verwenden.

Definition 1.2. Let $\varphi, \psi : \mathcal{M} \to \mathbb{C}$ be positive lineare functionals over a von Neumann algebra $\mathcal{M}$ such that $\varphi \ll\gg \psi$ was das nochmal genau??? and $f : (0, \infty) \to \mathbb{R}$ be a convex function with $f(1) = 0$ (brauche ich das hier???). The *f-divergence* of φ and ψ is given by

$$\begin{aligned} D_f(\varphi \mid \psi) &= \int_1^\infty f''(\gamma) E_\gamma(\varphi \mid \psi) + \gamma^{-3} f''(\gamma^{-1}) E_\gamma(\psi \mid \varphi) d\gamma \\ &= \int_1^\infty f''(\gamma) E_\gamma(\varphi \mid \psi) + \int_0^1 \gamma f''(\gamma) E_{\frac{1}{\gamma}}(\psi \mid \gamma) d\gamma. \end{aligned} \tag{1}$$

1.2 First Inequalitie

In [SV16] there are a lot of inequalities for f-divergences in the context of probability measures.

We want to generalize them for states over arbitrary von Neumann algebras (even type vNA of type III).

We start with inequalitie (84) from [SV16].

Proposition 1.3. Let $f, g : (0, \infty) \to \mathbb{R}$ convex functions with $f(1) = g(1)$, $f'(1) = g'(1)$ and $f(t) \leq g(t)$ for all $t \in (0, \infty)$. Then, the following inequalitie holds:

$$D_f(\varphi \mid \psi) \leq D_g(\varphi \mid \psi). \tag{2}$$

φ and ψ are states over an arbitrary von Neumann algebra.

Remark. Alexandrovs theorem states that every convex function defined on an open domain is twice differentiable almost everywhere.

Remark. By assumption $f'(1) = g'(1)$ indirectly require, that f and g are differentiable at $x = 1$.

2 Section 2

2.1 Subsection A

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References

[SV16] Igal Sason and Sergio Verdu. "f -Divergence Inequalities". In: *IEEE Transactions on Information Theory* 62.11 (Nov. 2016), pp. 5973–6006. ISSN: 1557-9654. DOI: 10.1109/tit.2016.2603151. URL: http://dx.doi.org/10.1109/TIT.2016.2603151.